PROJECT REPORT

Task 5: Smart File Organizer
Synent Technologies – Python Development Internship
Developer: Harsh Bhanarkar

1. Introduction

Managing files manually becomes difficult as the number of downloaded documents, images, videos, and other files increases. This project automates file organization by categorizing files based on their extensions and moving them into appropriate folders.

2. Objective

• Automate file organization.
• Categorize files by extension.
• Reduce manual effort.
• Demonstrate Python file handling and modular programming.

3. Problem Statement

Users often keep all downloaded files in a single directory, making it difficult to locate important documents. The Smart File Organizer solves this by automatically sorting files into categorized folders.

4. Technologies Used

Programming Language: Python 3
Libraries: os, shutil, pathlib
Development Environment: Visual Studio Code
Version Control: Git & GitHub

5. System Design

The application scans the selected directory, identifies file extensions, creates destination folders if they do not already exist, and moves each file to the corresponding category while skipping unsupported or already organized files.

6. Workflow

1. Launch the application.
2. Read files from the target directory.
3. Identify each file extension.
4. Match the extension with a predefined category.
5. Create destination folders if required.
6. Move files into categorized folders.
7. Display a summary of the organization process.

7. Features Implemented

- Automatic file categorization
- Folder creation
- Support for multiple file types
- Organized output directory
- Modular code structure
- Basic exception handling

8. Folder Structure

assets/
docs/
Test_Files/
Organized_Files/
organizer.py
README.md
requirements.txt

9. Testing and Validation

The application was tested using sample PDF, DOCX, TXT, JPG, PNG, MP3, MP4 and ZIP files. It correctly categorized supported file types and created folders automatically when necessary.

10. Results

The application successfully organized mixed files into their respective folders, reducing manual effort and improving directory management.

11. Future Enhancements

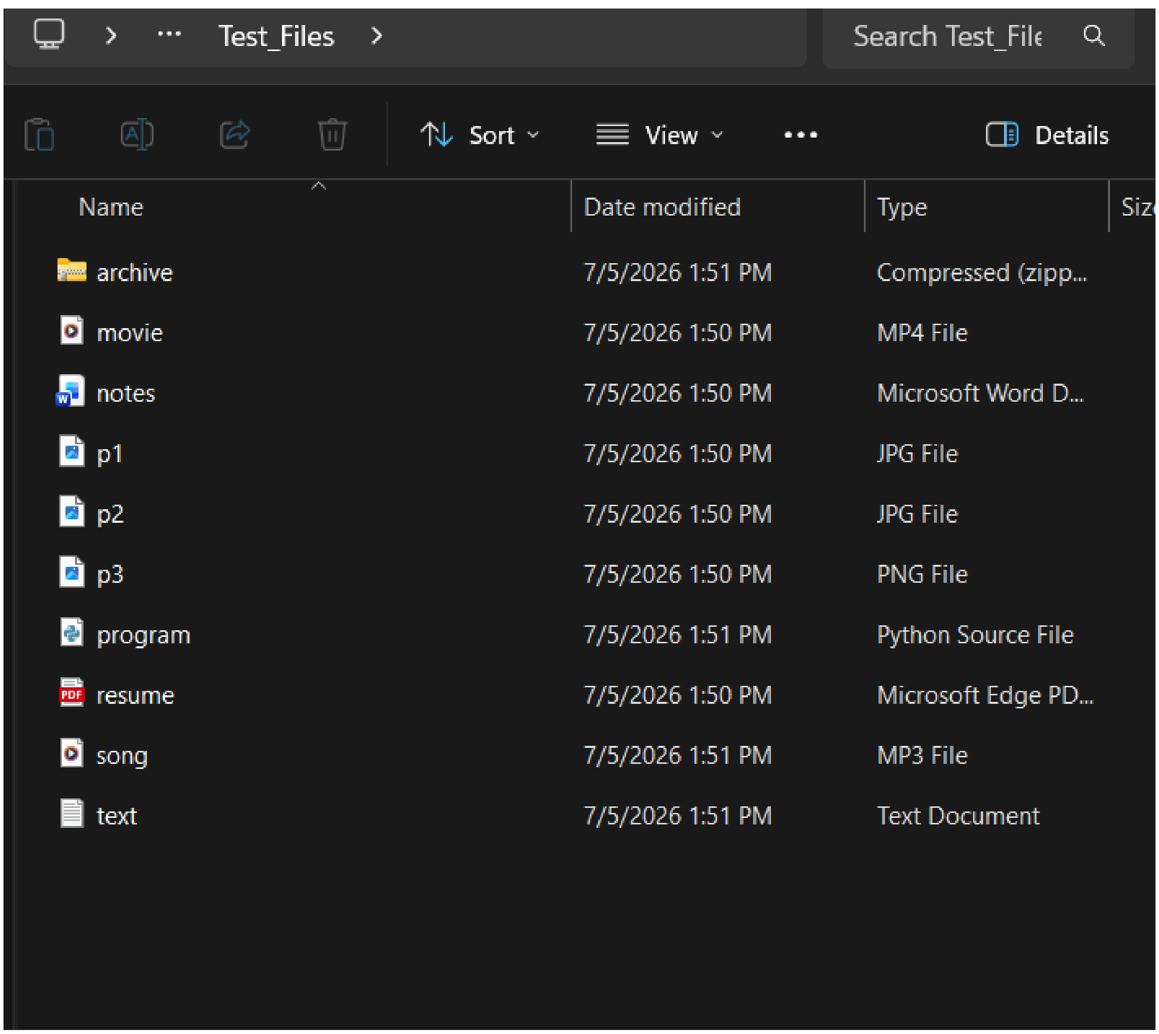
• Recursive folder organization
• Duplicate file detection
• GUI using Tkinter or PyQt
• Automatic scheduled organization
• User-defined file categories

12. Conclusion

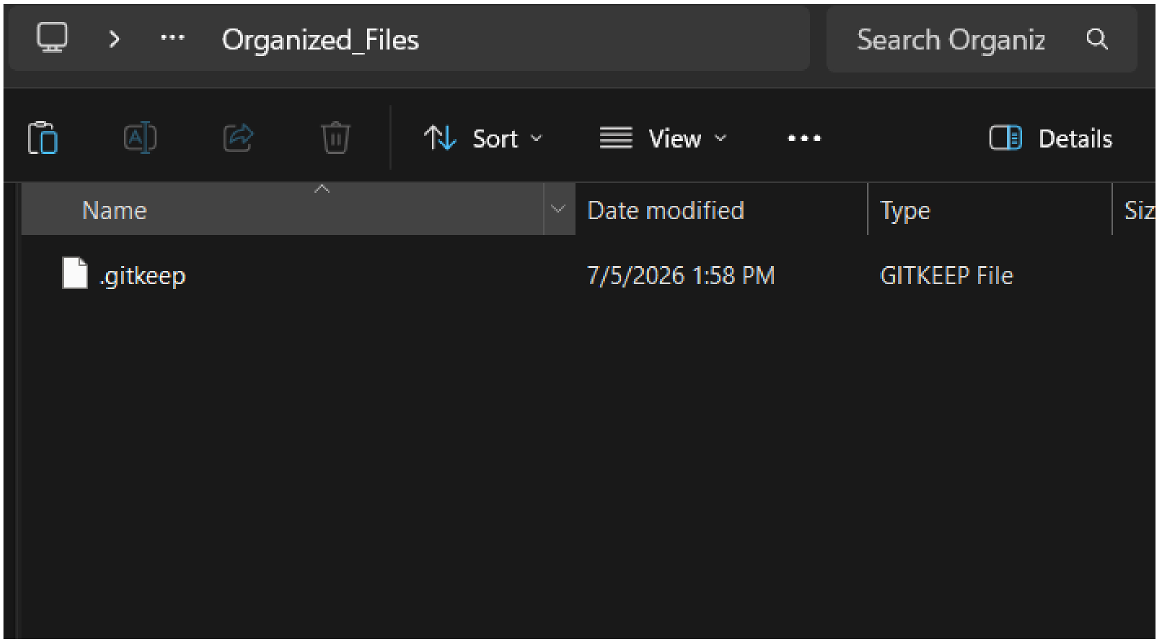
The Smart File Organizer demonstrates practical use of Python for automation through file handling, directory management, and modular software design. It provides a simple yet effective solution for organizing files.

Screenshots

Test_Files Folder:



Initial Organized_Files:



Updated Organized_Files